Network Traffic Analysis Report

General Information

Report Title

Network Traffic Analysis Report - Uneeq Interns Task

Analysis Date

May 5, 2026

Analyst

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Platform & Tools

- Kali Linux with Wireshark (Interface: wlan0)
- Web Browser
- Command Prompt

Objective

This analysis will:

- Capture and examine network traffic with Wireshark
- Identify security threats, abnormal behavior, and suspicious activity
- Document anomalies with clear, actionable observations
- Provide recommendations to harden the network and establish a baseline for normal traffic

Methodology: Traffic Capture & Generation

1

Step 1: Capture Traffic

Capture Duration: 10 minutes

Capture Method: Live capture via wlan0 interface

Filter Applied: Host 192.168.1.0/24

2

Step 2: Traffic Generation

- Open www.google.com (via Web Browser)
- Open www.youtube.com (via Web Browser)
- Open Google and Search for "Freepik"
- Open www.magnific.com (via Web Browser)
- Open Terminal and ping google.com

Methodology: Filters & Analysis

1

DNS Filter

dns: Captured all DNS queries and responses

2

TCP Filter

tcp: Filtered TCP protocol traffic

3

HTTP Filter

http: Isolated HTTP requests and responses

4

Additional Filters

tcp.analysis.flags: tcp errors

tls.handshake

5

Statistics

- Packet Hierarchy
- Endpoints
- Conversations
- I/O Graphs

6

Suspicious

- Follow TCP Stream of packets
- Check for random domains
- Export Object of HTTP payloads
- Check for errors and suspicious behavior

Analysis & Findings

✅ Normal Traffic Observed

- Network Traffic appears normal
  - Packet Lengths: 22848 Packets
  - Top Talker: 192.168.1.173 (Used Device)
  - HTTP Request Count: 89

- No suspicious IPs or domains detected

- No malicious payloads found

- All communications follow expected protocols

- All traffic is detected to known external servers over port 443

- Observed TCP reset (RST) packets during HTTPS communications

- TLS encrypted alerts detected, indicating normal session termination or handshake issue

Security Assessment

The analyzed network traffic indicates a **low-risk security posture** during the capture period. All observed communications were conducted over secure channels (HTTPS – port 443), targeting well-known and trusted external servers.

The presence of TCP reset (RST) packets and TLS encrypted alerts is consistent with normal network behavior, typically associated with session termination, timeout events, or minor handshake interruptions. No indicators of compromise (IoCs), such as suspicious domains, abnormal ports, or malicious payloads, were identified.

Overall, the traffic pattern reflects legitimate user activity without any evidence of exploitation attempts, data exfiltration, or command-and-control (C2) communication

Conclusion

Based on the analysis of the captured network traffic:

- No malicious or suspicious activity was detected
- All connections were established using standard and secure protocols
- Observed anomalies (RST packets and TLS alerts) fall within normal operational behavior

The network environment during the capture period can be considered **secure and operating as expected.**

Recommendations

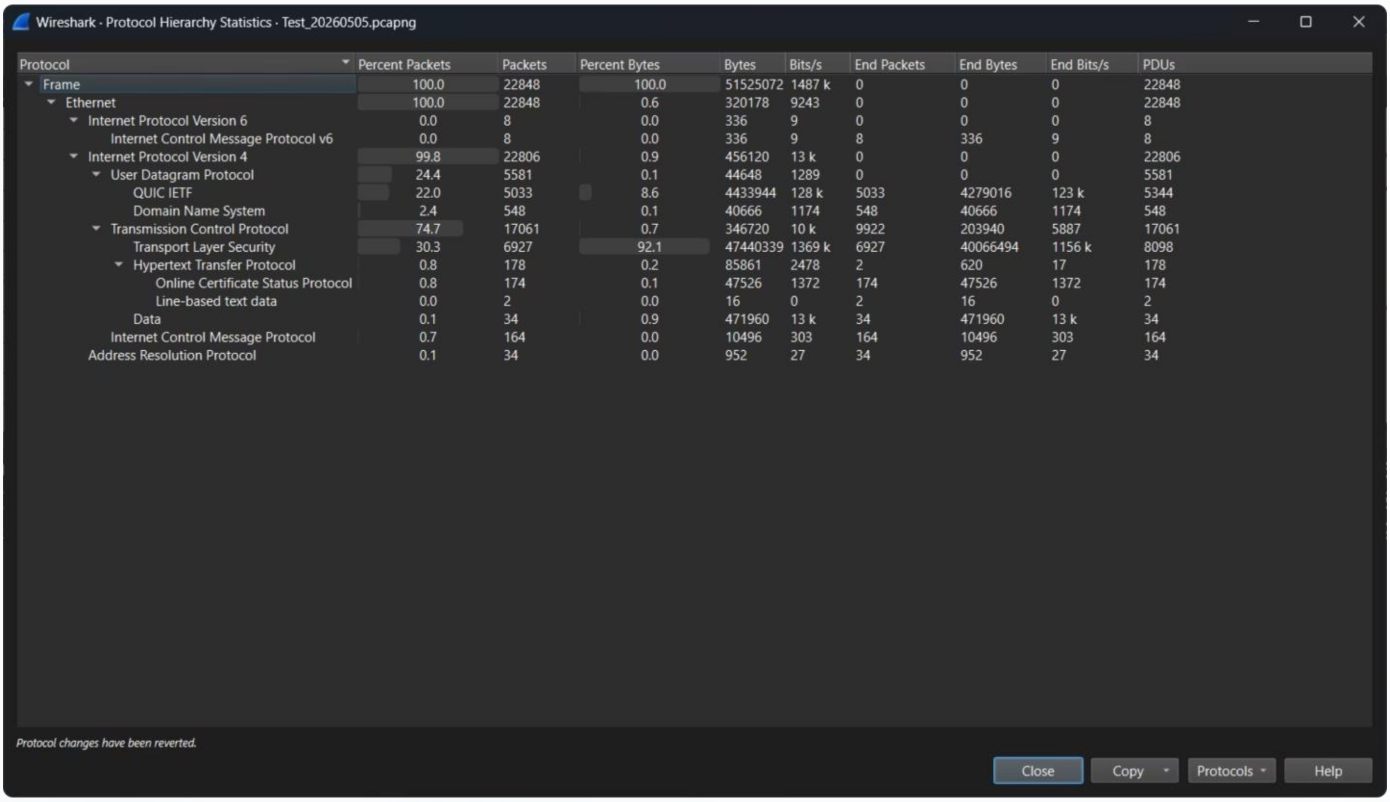
Although no threats were identified, the following best practices are recommended to maintain and enhance security:

- Continue monitoring network traffic regularly using tools like Wireshark
- Implement Intrusion Detection/Prevention Systems (IDS/IPS) for real-time threat detection
- Ensure all systems and applications are regularly updated and patched
- Use firewall rules to restrict unnecessary outbound/inbound traffic
- Enable logging and alerting mechanisms for unusual connection patterns (e.g., excessive RST packets)
- Conduct periodic security assessments and traffic analysis

Appendix

PCAP File Name: Test_20260505.pcapng

Supplementary Screenshots:


Wireshark · Protocol Hierarchy Statistics · Test_20260505.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	22848	100.0	51525072	1487 k	0	0	0	22848
Ethernet	100.0	22848	0.6	320178	9243	0	0	0	22848
Internet Protocol Version 6	0.0	8	0.0	336	9	0	0	0	8
Internet Control Message Protocol v6	0.0	8	0.0	336	9	8	336	9	8
Internet Protocol Version 4	99.8	22806	0.9	456120	13 k	0	0	0	22806
User Datagram Protocol	24.4	5581	0.1	44648	1289	0	0	0	5581
QUIC IETF	22.0	5033	8.6	4433944	128 k	5033	4279016	123 k	5344
Domain Name System	2.4	548	0.1	40666	1174	548	40666	1174	548
Transmission Control Protocol	74.7	17061	0.7	346720	10 k	9922	203940	5887	17061
Transport Layer Security	30.3	6927	92.1	47440339	1369 k	6927	40066494	1156 k	8098
Hypertext Transfer Protocol	0.8	178	0.2	85861	2478	2	620	17	178
Online Certificate Status Protocol	0.8	174	0.1	47526	1372	174	47526	1372	174
Line-based text data	0.0	2	0.0	16	0	2	16	0	2
Data	0.1	34	0.9	471960	13 k	34	471960	13 k	34
Internet Control Message Protocol	0.7	164	0.0	10496	303	164	10496	303	164
Address Resolution Protocol	0.1	34	0.0	952	27	34	952	27	34

Protocol changes have been reverted.

Close Copy Protocols Help

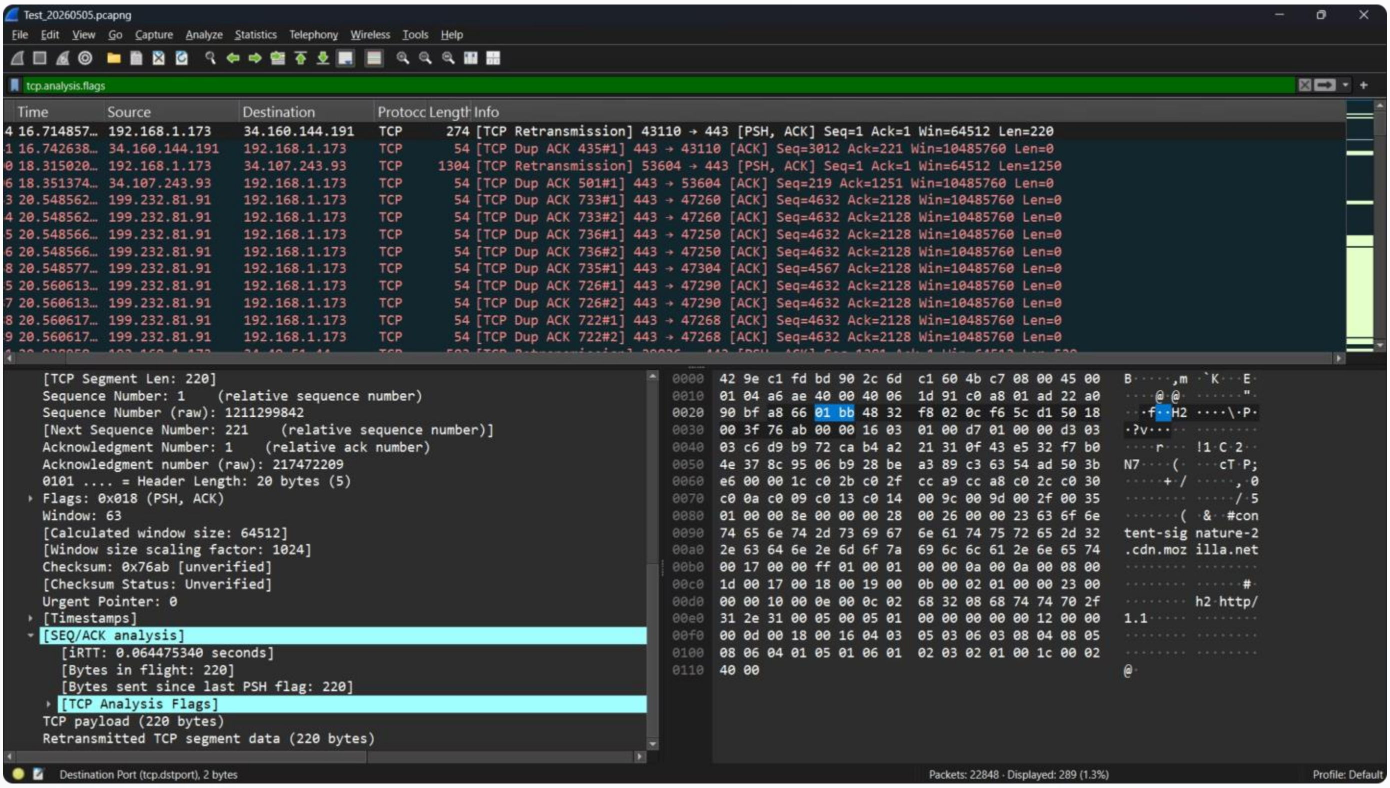

Wireshark · Expert Information · Test_20260505.pcapng

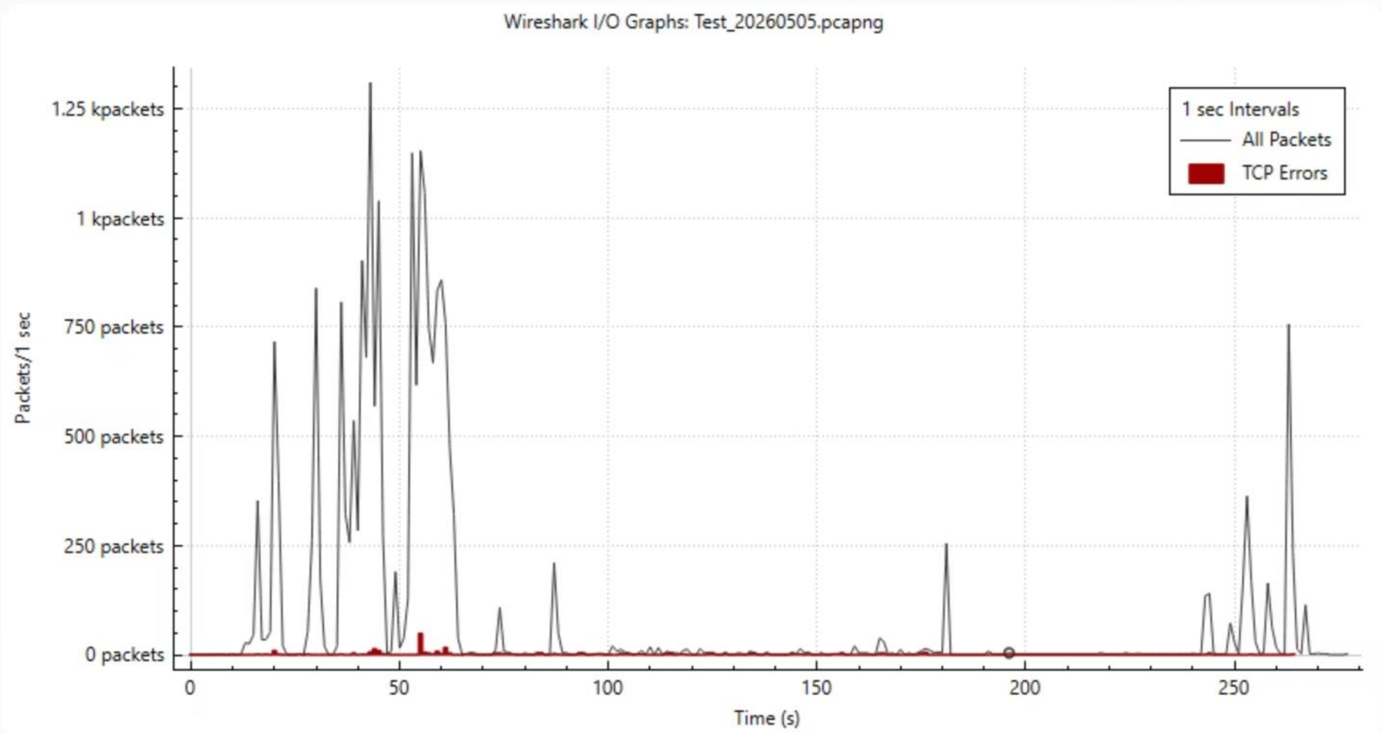
Severity	Summary	Group	Protocol	Count
Warning	Previous segment(s) not captured (common at capture start)	Sequence	TCP	5
Warning	This frame is a (suspected) out-of-order segment	Sequence	TCP	38
Warning	D-SACK Sequence	Sequence	TCP	8
Warning	Connection reset (RST)	Sequence	TCP	111
Warning	Failed to decrypt handshake	Decryption	QUIC	470
Note	The acknowledgment number field is nonzero while the ACK flag is not set	Protocol	TCP	3
Note	This frame is a (suspected) spurious retransmission	Sequence	TCP	6
Note	This frame undergoes the connection closing	Sequence	TCP	121
Note	This QUIC frame has a reused stream offset (retransmission?)	Sequence	QUIC	26
Note	ACK to a TCP keep-alive segment	Sequence	TCP	68
Note	TCP keep-alive segment	Sequence	TCP	68
Note	This frame initiates the connection closing	Sequence	TCP	161
Note	Coalesced Padding Data	Protocol	QUIC	156
Note	Duplicate ACK	Sequence	TCP	57
Note	This frame is a (suspected) retransmission	Sequence	TCP	51
Note	Partial Acknowledgement of a segment	Sequence	TCP	138
Chat	TCP window update	Sequence	TCP	2
Chat	Connection finish (FIN)	Sequence	TCP	282
Chat	This legacy_version field MUST be ignored. The supported_versions exte...	Deprecated	TLS	382
Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP	177
Chat	Connection establish request (SYN)	Sequence	TCP	164

No display filter set.

Limit to Display Filter Group by summary Search: Show...

Close Help


Test_20260505.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.analysis.flags
Time Source Destination Protocol Length Info
16.714857… 192.168.1.173 34.160.144.191 TCP 274 [TCP Retransmission] 43110 → 443 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=220
16.742638… 34.160.144.191 192.168.1.173 TCP 54 [TCP Dup ACK 435#1] 443 → 43110 [ACK] Seq=3012 Ack=221 Win=10485760 Len=0
18.315020… 192.168.1.173 34.107.243.93 TCP 1304 [TCP Retransmission] 53604 → 443 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=1250
18.351374… 34.107.243.93 192.168.1.173 TCP 54 [TCP Dup ACK 501#1] 443 → 53604 [ACK] Seq=219 Ack=1251 Win=10485760 Len=0
20.548562… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 733#1] 443 → 47260 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.548562… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 733#2] 443 → 47260 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.548566… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 736#1] 443 → 47250 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.548566… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 736#2] 443 → 47250 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.548577… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 735#1] 443 → 47304 [ACK] Seq=4567 Ack=2128 Win=10485760 Len=0
20.560613… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 726#1] 443 → 47290 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.560613… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 726#2] 443 → 47290 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.560617… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 722#1] 443 → 47268 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
20.560617… 199.232.81.91 192.168.1.173 TCP 54 [TCP Dup ACK 722#2] 443 → 47268 [ACK] Seq=4632 Ack=2128 Win=10485760 Len=0
[TCP Segment Len: 220]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1211299842
[Next Sequence Number: 221 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 217472209
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 63
[Calculated window size: 64512]
[Window size scaling factor: 1024]
Checksum: 0x76ab [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[iRTT: 0.064475340 seconds]
[Bytes in flight: 220]
[Bytes sent since last PSH flag: 220]
[TCP Analysis Flags]
TCP payload (220 bytes)
Retransmitted TCP segment data (220 bytes)
Destination Port (tcp.dstport), 2 bytes
Packets: 22848 · Displayed: 289 (1.3%)
Profile: Default


Wireshark I/O Graphs: Test_20260505.pcapng
1.25 kpackets
1 kpackets
750 packets
500 packets
250 packets
0 packets
Packets/1 sec
0
50
100
150
200
250
Time (s)
1 sec Intervals
All Packets
TCP Errors